

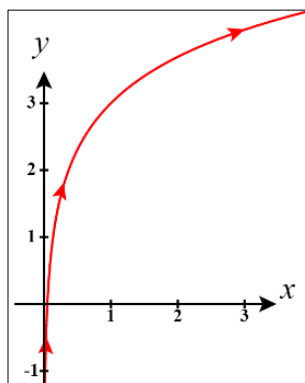
Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Consider the curve $x(t) = e^t$, $y = t + 3$. Eliminate the parameter to find a Cartesian equation of the curve. Then sketch the curve and indicate the orientation of the curve.

Note that $x = e^t \rightarrow \ln(x) = t$. Therefore, $y = t + 3 \rightarrow y = \ln(x) + 3$.

So, this is the graph of $y = \ln(x)$ shifted up 3 units.



2. Let C be the portion of curve of $y = \sqrt{x}$ from $(1,1)$ to $(4,2)$.
a) Create two different parameterizations for C .

Let $x(t) = t$ and $y = \sqrt{t}$ where $1 \leq t \leq 4$.

Also $x(t) = t^2$ and $y = t$ where $1 \leq t \leq 2$.

- b) Create a parameterization for $-C$ (this is the same as the curve C but traced in the opposite direction).

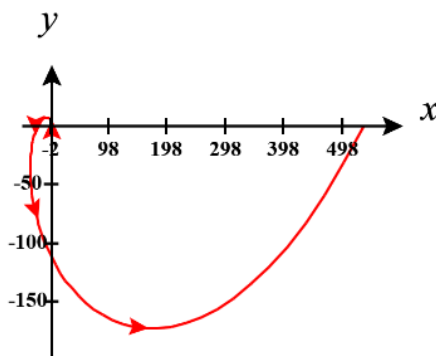
Let $x(t) = 5 - t$ and $y = \sqrt{5 - t}$ where $1 \leq t \leq 4$.

Note that $x(t)$ reverses the inputs for values of t .

3. Consider the curve $x = e^t \cos(t)$, $y = e^t \sin(t)$.

a) Make a sketch of the curve for $t \geq 0$ by plotting points. Don't use your calculator.

t	x	y
0	1	0
$\pi/2$	0	1
π	$-e^\pi$	0
$3\pi/2$	0	$-e^{3\pi/2}$
2π	$e^{2\pi}$	0



b) Determine when the tangent line to this equation is horizontal and when it is vertical.

Note $x'(t) = e^t \cos(t) - e^t \sin(t)$ & $y'(t) = e^t \sin(t) + e^t \cos(t)$.

Therefore, $x'(t) = 0 \rightarrow e^t [\cos(t) - \sin(t)] = 0 \rightarrow \cos(t) = \sin(t) \rightarrow t = \frac{\pi}{4} + \pi k$ and

$y'(t) = 0 \rightarrow e^t [\sin(t) + \cos(t)] = 0 \rightarrow \sin(t) = -\cos(t) \rightarrow t = \frac{3\pi}{4} + \pi k$ where k is some integer.

This implies that when $t = \frac{\pi}{4} + \pi k$ there are vertical tangent lines and when $t = \frac{3\pi}{4} + \pi k$ there are horizontal tangent lines.

c) Find the arc length for this path for t from 0 to 1.

We already calculated $x'(t)$ and $y'(t)$ in part (b) and see that these functions are continuous and since e^t is always increasing $x(t)$ and $y(t)$ never repeat coordinates. Therefore, the length of the curve is

$$\begin{aligned}
 L &= \int_0^1 \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt = \int_0^1 \sqrt{(e^t [\cos(t) - \sin(t)])^2 + (e^t [\sin(t) + \cos(t)])^2} dt \\
 &= \int_0^1 \sqrt{e^{2t} [\cos^2(t) - 2\cos(t)\sin(t) + \sin^2(t)] + e^{2t} [\sin^2(t) + 2\sin(t)\cos(t) + \cos^2(t)]} dt \\
 &= \int_0^1 \sqrt{2e^{2t}} dt = \int_0^1 \sqrt{2}e^t dt = \sqrt{2}e^t \Big|_0^1 = \sqrt{2}e - \sqrt{2}
 \end{aligned}$$

4. Consider the curve $x(t) = 2t^2$, $y(t) = t^3 - 4t$.

a) Determine at what value(s) of t the curve reaches the point $(8, 0)$.

In order to arrive at the point $(1, 0)$ we need to find a value of t that makes the x formula output 1 and the y formula output 0.

$$\text{We see that } 2t^2 = 8 \rightarrow t^2 = 4 \rightarrow t = \pm 2$$

$$\text{Also } t^3 - 4t = 0 \rightarrow t(t^2 - 4) = 0 \rightarrow t(t-2)(t+2) = 0 \rightarrow t = 0 \text{ or } t = \pm 2.$$

Therefore, the point $(8, 0)$ is reached when $t = -2$ and when $t = 2$.

b) Find all values of t for which the tangent line is horizontal or is vertical.

$$\text{Note } x'(t) = 4t \text{ \& } y'(t) = 3t^2 - 4.$$

$$\text{Therefore, } x'(t) = 0 \rightarrow 4t = 0 \rightarrow t = 0 \text{ and } y'(t) = 0 \rightarrow 3t^2 - 4 = 0 \rightarrow t = \pm \frac{2}{\sqrt{3}}.$$

This implies that when $t = 0$ (at the point $(0, 0)$) there is a vertical tangent line and when $t = \pm \frac{2}{\sqrt{3}}$ (at

the points $\left(\frac{8}{3}, \frac{16}{3\sqrt{3}}\right)$ and $\left(\frac{8}{3}, -\frac{16}{3\sqrt{3}}\right)$) there are horizontal tangent lines.

c) Use $x'(t)$ and $y'(t)$ together with intercepts and behavior as $t \rightarrow \pm\infty$ to sketch the curve. Don't use your calculator.

$$\text{Note that } x(t) = 0 \rightarrow t = 0 \text{ and}$$

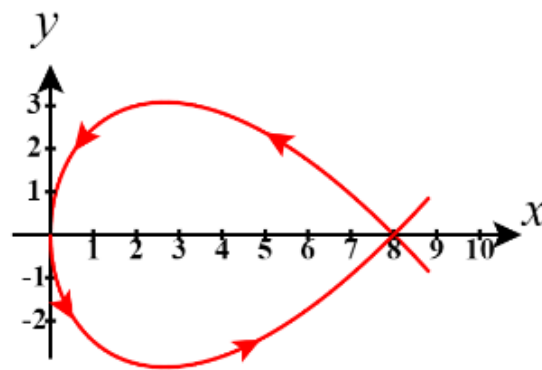
$$y(t) = 0 \rightarrow t^3 - 4t = 0 \rightarrow t(t^2 - 4) = 0 \rightarrow t = 0, \pm 2$$

So we see that the point $(0, 0)$ is on the graph when $t = 0$.

Also when $t = \pm 2$ we have the point $(8, 0)$.

We also note that as $t \rightarrow \pm\infty$ we have $x(t) \rightarrow \infty$ and

$$y(t) \rightarrow \pm\infty.$$



d) Find the slope of the two tangent lines at $(8, 0)$.

$$\text{From part (a) we know that } \frac{dy}{dx} = \frac{3t^2 - 4}{4t}.$$

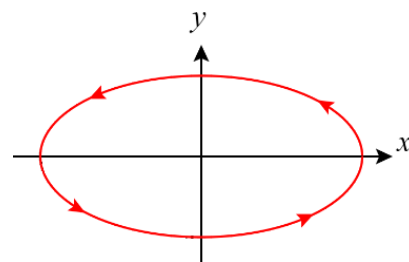
From part (b) we know that the point $(8, 0)$ is reached when $t = \pm 2$.

$$\text{Therefore, the slope of the two tangent lines at this point are } \left. \frac{dy}{dx} \right|_{t=-2} = \frac{12-4}{-8} = -1 \text{ and } \left. \frac{dy}{dx} \right|_{t=2} = \frac{12-4}{8} = 1$$

5. Consider the curve $x = a \cos(t)$, $y = b \sin(t)$.

- a) Show that this curve models a particle traveling counterclockwise about an ellipse starting at the point $(a, 0)$. Recall that the general form for the equation of an ellipse centered at the origin is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Note that $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \cos^2(t) + \sin^2(t) = 1$ and we know this is the equation of an ellipse. When $t = 0$ we are at the point $(a, 0)$ and at $t = \frac{\pi}{2}$ the point $(0, b)$.



- b) Determine for what t the tangent line has slope $\frac{a}{b}$.

We calculate $\frac{dy}{dx} = \frac{b \cos(t)}{-a \sin(t)} = -\frac{b}{a} \cot(t)$.

Therefore $-\frac{b}{a} \cot(t) = \frac{a}{b} \rightarrow -\frac{b^2}{a^2} = \tan(t) \rightarrow t = \arctan\left(-\frac{b^2}{a^2}\right)$.

- c) Setup, but do not evaluate, the integral for the circumference of the ellipse using this set of parametric equations. This integral is an “elliptical” integral and is another integral that does not have a nice anti-derivative.

Note that on the interval $[0, 2\pi]$, $x'(t)$ & $y'(t)$ are continuous and the curve is not retraced.

Therefore, $L = \int_0^{2\pi} \sqrt{a^2 \cos^2(t) + b^2 \sin^2(t)} dt$.

6. The parameterization $x = \cos(t)$, $y = \sin(t)$ for $0 \leq t \leq 2\pi$ creates a unit circle with counterclockwise orientation.

a) Create a parameterization for a unit circle with clockwise orientation.

$$\text{Let } x = \sin(t), y = \cos(t) \text{ for } 0 \leq t \leq 2\pi$$

b) Create a parameterization for a circle of radius r .

$$\text{Let } x = r \cos(t), y = r \sin(t) \text{ for } 0 \leq t \leq 2\pi$$

OR

$$\text{Let } x = r \sin(t), y = r \cos(t) \text{ for } 0 \leq t \leq 2\pi$$

c) Create a parameterization for the following ellipse.

$$\text{Let } x(t) = 2 \cos(t) \text{ \& } y(t) = 4 \sin(t) \text{ where } 0 \leq t \leq 2\pi.$$

